

1.FACTORIAL WITH ITERATIVE METHOD:

Iteration is a process of repeatedly executing a set of statements using a loop until a particular condition becomes false.

In factorial, we repeatedly multiply the number by the decreasing values until it reaches 1.

ALGORITHM:

1. Start.
2. Read a number n from the user.
3. Set fact = 1.
4. Check whether $n > 0$.
5. Multiply fact by n .
6. Decrease n by 1.
7. Repeat steps 4–6 until n becomes 0.
8. Display the factorial.
9. Stop.

Time Complexity:

Case	Time Complexity
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Best Case	$O(1)$
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Average Case	$O(n)$
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Worst Case	$O(n)$
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Space Complexity:

Space Complexity = $O(1)$

OUTPUT:

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Enter a number: 6
Factorial = 720

...Program finished with exit code 0
Press ENTER to exit console. 
```

CONCLUSION:

The factorial of a number was successfully calculated using an iterative approach with a while loop. The program repeatedly multiplies the number by decreasing values until it reaches zero. This method is simple and uses less memory, with **$O(n)$ time complexity** and **$O(1)$ space complexity**.